

A Comparative study of Lightweight Deep Learning Frameworks for Plant Disease Detection

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Abstract—Accurate and early diagnosis of plant diseases is essential for better crop yield production. With the growing adoption of mobile and edge computing solutions in precision agriculture, there has been a growing interest in the use of lightweight deep learning models for their less complexity and higher efficiency. In this paper, a comparative analysis of three different lightweight convolutional neural network models, namely SqueezeNet, MobileNetV2, and EfficientNet-B0, for plant disease diagnosis is proposed. Each of the models was trained and tested on the same plant disease image dataset to ensure a fair comparison. The performance of the models was tested mainly based on classification accuracy, with a focus on their appropriateness for mobile and edge computing platforms. The experimental outcomes show that SqueezeNet has the best accuracy of 98.11%, performing better than EfficientNet-B0 with an accuracy of 98.06% and MobileNetV2 with an accuracy of 92.97%.

Index Terms—Crop disease detection, Deep learning, CNN, SqueezeNet, MobileNetV2, EfficientNet-B0, Lightweight models

I. INTRODUCTION

Plant disease detection has emerged as a crucial use of computer vision in agriculture, with the potential to enhance crop sustainability, yield, and health through prompt and precise diagnosis. Although deep learning models have shown impressive results in the identification of plant diseases from leaf images in recent years, their high computational and memory requirements restrict their practical deployment on resource-constrained devices like smartphones, drones, and edge sensors frequently used in agricultural settings [1] [2]. This problem is addressed by lightweight deep learning frameworks, which offer effective architectures and optimized inference with respectable accuracy.

This paper presents a complete workflow for developing a robust crop disease detection system using light weight deep learning techniques. The system is designed to identify

plant diseases from a dataset of leaf images using CNN-based architectures. The work begins with dataset preparation, where images are resized, normalized, and augmented to ensure better learning and reduce overfitting. Three lightweight models : SqueezeNet, MobileNetV2, EfficientNet-B0 - form the core of the study.

Model performance is systematically evaluated by using standard metrics which includes accuracy, precision, recall, and F1-score.

II. LITERATURE STUDY

Researchers have studied lightweight deep learning architectures as an effective solution for plant disease detection while achieving high accuracy and maintaining low computational requirements. EfficientNet and its variant models use compound scaling methods to achieve high accuracy which enables their use in environments with limited resources. [3]

MobileNetV2 has become a standard for crop disease classification because its design includes depthwise separable convolutions which provide efficient performance for detecting tomato, apple diseases and diabetic foot ulcers. [4]

SqueezeNet-based methods have demonstrated successful results for detecting diseases in tomato, maize, apple and banana leaves while they achieve high classification accuracy through their ability to decrease model size. [5]

ShuffleNet and its improved variants have also been used for plant disease recognition and feature extraction because they provide quick inference times and require low processing power. [6]

Researchers have developed YOLO-based models like YOLO-Nano and Tiny-YOLO to enable agricultural disease detection through aerial imagery in real-time. [7]

The studies demonstrate that lightweight CNN frameworks can deliver accurate plant disease detection results which lead to efficient performance.

TABLE I
COMPARATIVE LITERATURE REVIEW OF CROP DISEASE DETECTION MODELS

Year	Title	Dataset	Model / Library	Findings	Accuracy	Limitations
2019	YOLO Nano: A Compact You Only Look Once CNN for Object Detection [8]	1. PASCAL VOC 2007/2012 2. Natural images 3. 20 object classes	1. YOLO Nano 2. Human-machine collaborative design 3. PEP architecture 4. FCA attention	1. Balances accuracy and size 2. Efficient embedded deployment	69.1% mAP (VOC 2007) Outperforms Tiny YOLOv2/v3	1. Existing models are large and slow 2. Tiny YOLO sacrifices accuracy
2020	EfficientNet: Rethinking Model Scaling for CNNs [9]	1. ImageNet dataset 2. CIFAR-100, Flowers 3. Transfer learning datasets	1. EfficientNet family 2. EfficientNet-B0 3. MBConv + squeeze-and-excitation	1. Studies compound model scaling 2. Balances depth, width, resolution	Top-1: 84.4% Top-5: 97.1% (ImageNet)	1. Manual tuning required 2. Depends on baseline network
2024	Tiny YOLO Networks for Disease Detection in Paddy Agronomy [10].	1. Rice leaf dataset 2. 15,456 images 3. 3 disease classes	1. Modified Tiny YOLOv4 2. SPP + attention modules 3. UAV-based detection	1. Detects small diseased regions 2. Improved UAV accuracy	mAP: 86.97%	1. Small infection points hard to detect 2. Accuracy limited despite real-time speed
2024	MC-ShuffleNetV2: A Lightweight Model for Maize Disease Recognition [11].	1. 2,725 field images 2. Augmented to 10,845 images	1. MC-ShuffleNetV2 2. CBAM attention 3. Mish activation	1. High accuracy 2. Very low FLOPs	Accuracy: 99.86%	1. Limited disease classes 2. Dataset size constraints
2025	SqueezeNet-Based DL Framework for Accurate Tomato Leaf Disease Diagnosis & Classification [12].	1. PlantVillage (Mendeley Data) 2. 14,531 RGB images 3. 9 diseases + healthy 4. Class imbalance	1. SqueezeNet architecture 2. Transfer learning (ImageNet) 3. SGDM, ADAM, RMSPROP 4. MATLAB and Python	1. Efficient deployable model 2. High accuracy with low complexity 3. Training parameter analysis	Overall: 99.91% Healthy recall: 100% F1: 99.42%	1. Multiple diseases on one leaf not handled 2. Limited real-world representation

III. DEEP LEARNING MODELS

The experimental study began with an analysis of five different convolutional neural network (CNN) models:

- 1) **YOLO** — YOLO (Tiny-YOLO or YOLO-Nano) can be applied for the detection of the disease patches on leaves.
 - The image of the crop leaf is divided into grids.
 - If a disease spot appears in a cell of the grid, a bounding box is created around it.
 - The generated vector represents:
 - Position of the diseased region on the leaf.
 - Type of disease.
 - Confidence score (how certain the model is).
 - In this manner, YOLO performs both localization and classification simultaneously, localization identifies where a disease exists while classification identifies what disease it is.
- 2) **EfficientNet-Lite** — EfficientNet-Lite is a scaled-down version of EfficientNet which is optimized for running on mobile and edge devices. It is lightweight, fast, and accurate, making it appropriate for real-time crop disease detection. A technique called compound scaling (scaling all 3 parameters in an efficient way) is used, which balances all the three key aspects of a neural network:
 - **Depth** – number of layers in the network.
 - **Width** – number of channels per layer.
 - **Resolution** – input image size.

By scaling all the three parameters together in a balanced way, EfficientNet-Lite achieves a very high accuracy with lesser parameters and lower computation cost as compared to others.

Layer-wise Working of EfficientNet

- **Input Image** – The farmer uploads a photo of a leaf. The image is resized.

- **Convolution Layers (Feature Extraction)** – Small filters slide over the image in this layer. Patterns like spots, edges, and color patches (predefined patterns) are detected.
 - Early layers → Detect simple features (edges, colors).
 - Later layers → Detect complex features (disease textures, leaf shapes).
- **Pooling Layers (Simplification)** – Reduce image size while keeping only the most important features.
- **Fully Connected Layers (Decision Making)** – Combine extracted features. Classify the image into possible crop diseases.
- **Output** – Produces probabilities for each class.

- 3) **SqueezeNet** — SqueezeNet is a lightweight convolutional neural network designed to help in achieving high accuracy with very small model size. It is especially suitable for mobile and edge devices which are used in smart agriculture, where memory and computation resources are limited. SqueezeNet uses a special building block called the Fire Module, which positively reduces the number of parameters and also maintains strong feature representation.

The architecture focuses more on replacing large convolution filters with smaller 1×1 and 3×3 filters which has made the network fast and efficient for all real-time crop disease detection.

Key Components of SqueezeNet:

- **Fire Module** – Core building block consisting of:
 - **Squeeze layer (1×1 convolutions)** – It helps in reducing the number of input channels.
 - **Expand layer (1×1 and 3×3 convolutions)** – It helps in learning richer features with fewer parameters.

Layer-wise Working of SqueezeNet

- **Input Image** – Farmer uploads a leaf image, which is resized to a fixed dimension (224×224) and normalized.
- **Convolution Layers (Initial Feature Extraction)** – Early convolution layers detect basic visual patterns such as edges, colors, and textures.
 - Early layers → simple features (edges, color variations)
 - Deeper layers → complex features (disease spots, texture changes, leaf damage)
- **Fire Modules (Efficient Feature Learning)** – Fire modules assist in the compressing and expanding features efficiently, which allows the network to learn disease patterns with very low memory usage.
- **Pooling Layers (Dimensionality Reduction)** – It reduces the spatial dimension while preserving the all important disease-related features and in improving the speed and efficiency of the model.
- **Final Convolution Layer (Classification Preparation)** – It maps extracted features to the number of crop disease classes.
- **Output Layer** – Output layer helps in giving the output by generating probability scores for each crop disease class and predicting the most likely disease.

4) **ShuffleNet** — ShuffleNet is a lightweight convolutional neural network designed for fast and efficient performance on mobile phones and low-power devices. It minimizes computation by using group convolutions along with channel shuffle operation, which allows information to flow effectively across all the feature channels. **Layer-wise Working of ShuffleNet:**

- **Input Image** – The leaf image is resized and normalized before being fed into the network.
- **Grouped Convolution Layers** – Convolutions are divided into groups in order to reduce the computational cost while extracting important visual features.
- **Channel Shuffle Operation** – Channels are shuffled to ensure information exchange between groups, improving feature learning.
- **Depthwise Convolutions** – Extract spatial features efficiently with low computational cost.
- **Pooling Layers** – Reduce feature map size while preserving disease-related information.
- **Output Layer** – Produces probability scores for each crop disease class and outputs the final prediction.

5) **MobileNet** — MobileNet is a lightweight convolutional neural network architecture specifically created for edge, mobile, and embedded devices. It focuses on striking a good balance between speed, accuracy, and computational efficiency, which makes it ideal for real-time crop disease diagnosis in settings with limited resources.

Depthwise separable convolutions, which significantly reduce the amount of parameters and computations compared to ordinary convolutions, are the fundamental component of MobileNet

By factorizing standard convolutions in this way, MobileNet achieves comparable accuracy with significantly lower model size and inference cost.

Layer-wise Working of MobileNet

- **Input Image** – The farmer uploads a leaf image, which is resized to a fixed resolution suitable for the network.
- **Convolution Layers (Feature Extraction)** – Small depthwise filters slide over the image to extract visual patterns.
 - Early layers → capture basic features such as edges, color gradients, and contours.
 - Deeper layers → learn complex disease-specific patterns like lesions, texture irregularities, and discoloration.
- **Depthwise Separable Convolutions** – These layers reduce redundancy by isolating spatial and channel-wise learning, allowing efficient feature extraction without sacrificing representational power.
- **Pooling / Global Average Pooling (Simplification)** – Spatial dimensions are reduced while preserving the most informative features, lowering overfitting risk and computational load.
- **Fully Connected Layers (Decision Making)** – The extracted features are aggregated and mapped to disease categories through dense layers.
- **Output** – The model produces probability scores for each disease class, indicating the likelihood of different crop health conditions.

Further, three of these models were implemented, and their architectural and experimental details are given below.

IV. PROPOSED METHODOLOGY

The proposed methodology presents a lightweight deep learning-based framework for automated crop disease detection using leaf images. The overall workflow consists of dataset preparation, model selection, training, inference, and performance evaluation.

A. Dataset Acquisition and Preparation

Three datasets were Studied:

- 1) PlantVillage Dataset
- 2) Tomato Leaf Disease Dataset
- 3) New Plant Diseases Dataset.

The above listed datasets were studied and the New Plant Diseases Dataset was chosen due to its greater diversity and broader disease coverage as compared to the other datasets.

Preprocessing included image resizing, normalization, data augmentation, and dataset splitting

Table II shows the dataset description of Plant disease dataset used in the presented study.

TABLE II
DATASET DESCRIPTION

Plant Village	No of images: Training set: ~70,000 images Validation set: ~18,000 images Total: ~88,000 images
	Categories: 38 <i>Description:</i> Covers major crops (apple, corn, grape, potato, tomato) and includes a diverse range of fungal, bacterial, and viral diseases, along with a "healthy" control class for each crop type.

B. Lightweight CNN Model Selection

The goal was to ensure deployability on low-resource devices, hence the methodology employs lightweight CNN architectures instead of computationally heavy networks. This makes sure that the models can run on devices like mobile devices serving the purpose for small farmers.

Models studied include EfficientNet, Tiny YOLO, SqueezeNet, MobileNet, and ShuffleNet. While multiple architectures such as YOLO and ShuffleNet were studied, but the three models were chosen because they provide the best balance between accuracy, computational efficiency, and model size, which aligns with the objective of deploying plant disease detection systems on mobile and edge devices.

The selected models are:

- **SqueezeNet** – Extremely compact architecture that uses *Fire modules* for high parameter efficiency. SqueezeNet well-suited for resource-constrained environments. [13]
- **MobileNetV2** – Employs *depthwise separable convolutions* to achieve fast inference and low power consumption. MobileNetV2 ensure stable learning behavior throughout training. [14]
- **EfficientNet-B0** – Utilizes *compound scaling* to balance network depth, width, and input resolution efficiently. The lightweight nature of EfficientNet-B0 ensures suitability for real-time and edge-based deployment. [15]

C. Model Training Strategy

Training was performed using a **batch size of 32, 5 and 25 epochs**, early stopping, and model checkpoints. Metrics include accuracy, precision, recall, and F1-score. Mixed-precision training was applied. Figure 1 illustrates the complete workflow of the proposed plant disease detection system.

D. Performance Evaluation

Performance was evaluated using **accuracy, precision, recall, F1-score**, and confusion matrices for class-wise analysis.

Developing a Plant Disease Classification Model



Fig. 1. Plant Disease Workflow

V. RESULTS AND DISCUSSION

To evaluate the learning progression and overall performance of the selected lightweight models, experimentation was conducted over two different training durations: 5 epochs and 25 epochs. The model performance was systematically evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. The results of these experiments are summarized in Table III and Table IV.

TABLE III
PERFORMANCE COMPARISON OF MODELS (5 EPOCHS)

Model	Accuracy	Precision	Recall	F1-score
SqueezeNet	87.00%	88.74%	87.10%	87.12%
EfficientNet-B0	91.76%	92.31%	91.76%	91.67%
MobileNetV2	86.59%	87.21%	86.59%	86.53%

As observed in the tables, extending the training from 5 to 25 epochs significantly improved the predictive capabilities across all architectures. EfficientNet-B0 established a strong early baseline at 5 epochs (91.76% accuracy). However, at 25 epochs, SqueezeNet and EfficientNet-B0 both converged to highly accurate states, achieving 98.11% and 98.06% accuracy, respectively. MobileNetV2 also saw a substantial improve-

TABLE IV
PERFORMANCE COMPARISON OF MODELS (25 EPOCHS)

Model	Accuracy	Precision	Recall	F1-score
SqueezeNet	98.11%	98.21%	98.06%	98.12%
EfficientNet-B0	98.06%	98.10%	98.06%	98.06%
MobileNetV2	92.97%	91.88%	92.73%	92.38%

ment, reaching an accuracy of 92.97%. All of the models also reach a well balanced F1-score.

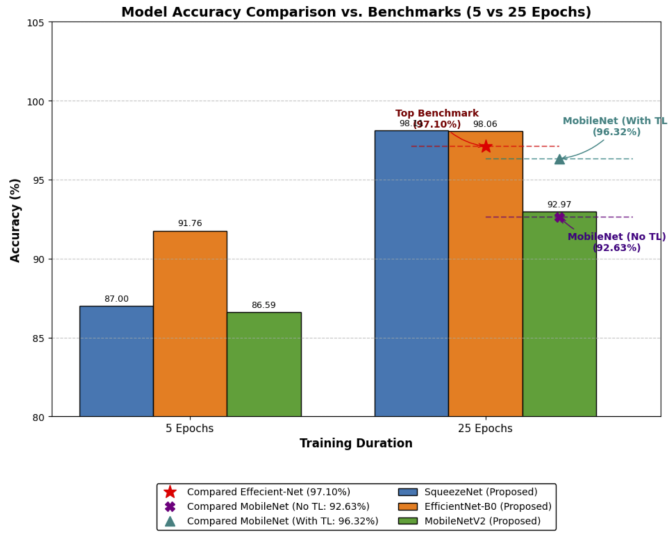


Fig. 2. Performance comparison

Figure 2 illustrates a performance comparison between our implemented models and existing benchmark architectures.

Furthermore, the experimental results obtained in this study surpass recent benchmarks in the field. When comparing our findings with the recent models presented in [16], our implementation yields strictly better accuracy and F1-scores; notably, our EfficientNet-B0 model achieved an accuracy of 98.06% compared to their reported 97.10%, representing an absolute improvement of 0.96%. While previous approaches often sacrifice precision for lower computational overhead, our optimized use of EfficientNet-B0 and SqueezeNet ensures superior disease classification rates while maintaining the lightweight profile necessary for agricultural edge deployment. Figure 2 is a graphical description for our model compared to the existing ones

To contextualize these findings, we compared the results with a recent study by Nigam et al. [17], which investigates deep learning models in uncontrolled agricultural environments. Since both studies evaluate MobileNet architectures, a direct comparison can be drawn: while their study reported a baseline accuracy of 92.63% for MobileNet-V1 without transfer learning (TL), our optimized MobileNetV2 achieves a superior accuracy of 92.97%. Although their MobileNet-

V1 model trained with transfer learning did achieve a higher accuracy than our MobileNetV2 but our proposed EfficientNet-B0 (98.06%) and SqueezeNet (98.11%) models perform on par with, or slightly better than, the absolute top-performing models in their study that utilized a TL approach. These results prove that carefully scaled lightweight CNNs can deliver state-of-the-art reliability for agricultural edge deployment.

VI. CONCLUSION

This research work has provided a comparative assessment of three lightweight deep learning models, namely SqueezeNet, MobileNetV2, and EfficientNet-B0, for plant disease diagnosis in terms of accuracy and feasibility of implementation in resource-limited settings. All three models have shown satisfactory results, thus validating the use of lightweight convolutional neural networks for automatic plant disease classification. Of the three models assessed at 25 epochs, SqueezeNet and EfficientNet-B0 achieved nearly identical classification accuracies of 98.11% and 98.06% respectively, both outperforming MobileNetV2 (92.97%). While the performance difference between SqueezeNet and EfficientNet-B0 is marginal, the overall results do bring out the significance of trade-offs between model complexity and accuracy. The results thus indicate that EfficientNet-B0 or SqueezeNet are the best models for practical implementation in plant disease detection systems, especially for mobile and edge-based agricultural applications. Future research work can be directed towards assessing the performance of other lightweight models, pruning or quantizing existing models, and testing their performance on real-world field data to improve accuracy and robustness.

VII. DECLARATIONS

- **Author contribution:** Ishaan Jain (IJ), Khushi Mittal (KM), and Namratha Reddy (NR) worked under the supervision of Dr. Jaspal Kaur Saini (JKS) for their Minor Project. This work was carried out at Dr. B. R. Ambedkar National Institute of Technology (NIT), Jalandhar. IJ, KM, and NR implemented the deep learning frameworks with insights of JKS. Three of them wrote the initial draft and JKS revised the final manuscript. The presented work is an outcome of their Minor Project for the degree of B.Tech IT.

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